

Tutorial 7-3

SUN YIFEI

sunyf2019@mail.sustc.edu.cn

P 7.47

7.47. Suppose $x[n]$ has a Fourier transform that is zero for $\pi/3 \leq |\omega| \leq \pi$. Show that

$$x[n] = \sum_{k=-\infty}^{\infty} x[3k] \left(\frac{\sin(\frac{\pi}{3}(n-3k))}{\frac{\pi}{3}(n-3k)} \right).$$

P 7.47

Let us define a signal

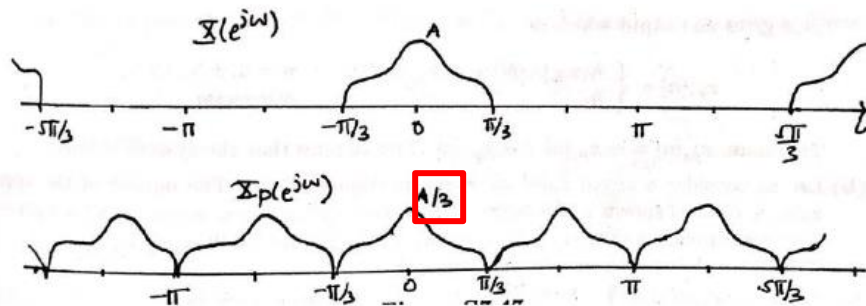
$$x_p[n] = x[n] \sum_{k=-\infty}^{\infty} \delta[n-3k] = \sum_{k=-\infty}^{\infty} x[3k] \delta[n-3k].$$

From Section 7.5.1, we know that the Fourier transform of $x_p[n]$ is

$$X_p(e^{j\omega}) = \frac{1}{3} \sum_{k=0}^2 X(e^{j(\omega-2\pi k/3)}).$$

Since $X_p(e^{j\omega}) = 0$ for $\pi/3 \leq |\omega| \leq \pi$, there is no aliasing among the replicas of $X(e^{j\omega})$ in $X_p(e^{j\omega})$.

This is shown below.



$$\frac{\sin Wn}{\pi n} = \frac{W}{\pi} \operatorname{sinc}\left(\frac{Wn}{\pi}\right)$$

$$0 < W < \pi$$

$$X(\omega) = \begin{cases} 1, & 0 \leq |\omega| \leq W \\ 0, & W < |\omega| \leq \pi \end{cases}$$

$X(\omega)$ periodic with period 2π

In order to be able to recover $x[n]$ from $x_p[n]$, it is clear that we need to pass $x_p[n]$ to a lowpass filter with cutoff frequency $\pi/3$ and passband gain 3. Therefore,

$$\begin{aligned} x[n] &= x_p[n] * \frac{3 \sin(\pi n/3)}{\pi n} \\ &= \left\{ \sum_{k=-\infty}^{\infty} x[3k] \delta[n-3k] \right\} * \frac{3 \sin(\pi n/3)}{\pi n} \\ &= \sum_{k=-\infty}^{\infty} x[3k] \frac{\sin(\pi(n-3k)/3)}{\pi(n-3k)/3} \end{aligned}$$

P 7.49

7.49. As discussed in Section 7.5 and illustrated in Figure 7.37, the procedure for interpolation or upsampling by an integer factor N can be thought of as the cascade of two operations. The first operation, involving system A , corresponds to inserting $N - 1$ zero-sequence values between each sequence value of $x[n]$, so that

$$x_p[n] = \begin{cases} x_d\left[\frac{n}{N}\right], & n = 0, \pm N, \pm 2N, \dots \\ 0, & \text{otherwise} \end{cases}$$

For exact band-limited interpolation, $H(e^{j\omega})$ is an ideal lowpass filter.

- (a) Determine whether or not system A is linear.
- (b) Determine whether or not system A is time invariant.
- (c) For $X_d(e^{j\omega})$ as sketched in Figure P7.49 and with $N = 3$, sketch $X_p(e^{j\omega})$.
- (d) For $N = 3$, $X_d(e^{j\omega})$ as in Figure P7.49, and $H(e^{j\omega})$ appropriately chosen for exact band-limited interpolation, sketch $X(e^{j\omega})$.

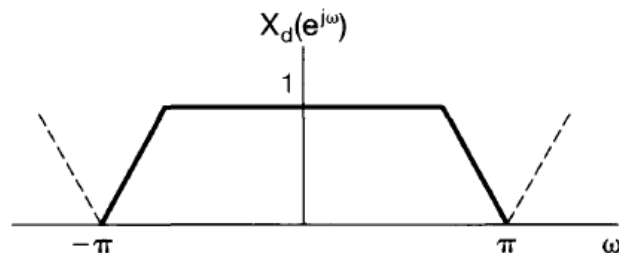


Figure P7.49

P 7.49

(a) Let the signals $x_{d_1}[n]$ and $x_{d_2}[n]$ be inputs to system A. Let the corresponding outputs be $x_{p_1}[n]$ and $x_{p_2}[n]$. Now consider an input of the form $x_{d_3}[n] = \alpha_1 x_{d_1}[n] + \alpha_2 x_{d_2}[n]$. This gives the output which is

$$x_{p_3}[n] = \begin{cases} \alpha_1 x_{d_1}[n/N] + \alpha_2 x_{d_2}[n/N], & n = 0, \pm N, \pm 2N, \dots \\ 0, & \text{otherwise} \end{cases}$$

Therefore, $x_{p_3}[n] = \alpha_1 x_{p_1}[n] + \alpha_2 x_{p_2}[n]$. Hence, the system is linear.

(b) Let us consider a signal $x_d[n]$ as shown in the figure below. The output of the system $x_p[n]$ is then as shown in the figure. Let us define a new input $x_{d_1}[n] = x_d[n-1]$. The corresponding $x_{p_1}[n]$ is shown in the figure. Clearly, $x_{p_1}[n] \neq x_p[n]$. Hence, the system is not time invariant.

(c) We have $X_p(e^{j\omega}) = X_d(e^{j\omega N})$. This is shown below.

(d) $X(e^{j\omega})$ is shown below.

